Smart-CALM: ACK-Aware Online Redundancy Control for Airtime-Constrained LoRa Mesh Networks

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***Abstract*—Low-power LoRa mesh networks extend coverage beyond a single radio hop, but each relay consumes long airtime on a shared and collision-prone channel. Flooding protects delivery at high channel cost, whereas cached source routes are efficient but can fail when links age or acknowledgments are lost. We present Smart-CALM, an ACK-aware online redundancy controller that selects among lean, balanced, and rescue forwarding profiles using a compact node-local tabular policy. The data plane retains explicit source-route discovery, bounded fallback forwarding, and reverse-path end-to-end ACKs. The controller uses recent delivery, route-cache, collision, and delay signals to adapt route lifetime, discovery timing, and timeout-rescue radius. In a packet-level LoRa simulation with 50 nodes and 20 common random seeds per scenario, Smart-CALM improves ACK-confirmed unicast delivery over flooding, source-route caching, and nonadaptive CALM in repeated-unicast, mixed-traffic, and stronger-shadowing scenarios. In mixed traffic it reaches 0.937 ACK PDR with 836.6 s of airtime, compared with 0.852 and 1182.9 s for managed flooding. The gain is not free: the adaptive method produces a longer tail ACK delay and spends more fallback forwarding than CALM. We therefore frame the result as a reliability-airtime-delay tradeoff and release the complete reproducible comparison matrix.**

***Index Terms*—LoRa mesh, low-power IoT, adaptive routing, online learning, ACK-aware reliability, airtime control**

I. Introduction

LoRa is attractive for sparse Internet-of-Things deployments because a low data rate can be exchanged for long range and low endpoint power [1]. A single-hop star, however, is not sufficient when terrain, buildings, or deployment cost prevent every sensor from reaching a gateway. Multihop and mesh extensions therefore remain an active design space [2]–[4]. Their central systems problem is that a relay improves reachability by consuming additional airtime on a channel where packets are long and overlapping transmissions are expensive.

Two common design points expose this tension. Managed flooding provides multiple forwarding opportunities and can preserve delivery when a single route is weak, but duplicate receptions and rebroadcasts increase channel occupancy. Route discovery followed by source-route caching reduces repeated control traffic, but a cached path is vulnerable to link variation, aging, and an ACK that is lost after the destination has already received DATA. Existing LoRa mesh work explores distance-vector routing, clustering, real-time forwarding, and implementation tradeoffs [5]–[9]; the remaining design question is how much redundancy to spend after the network has evidence that a path is becoming unreliable.

This paper presents Smart-CALM, a confidence-aware LoRa mesh controller that treats forwarding redundancy as a resource allocated from delivery evidence. The method does not require a global topology service or a neural model. A source first uses source-route forwarding. A compact node-local controller observes ACK-confirmed outcomes, route-cache behavior, collision pressure, and delay, then selects one of three prevalidated profiles. A delayed fallback transmission is scheduled only when the selected profile permits it; an ACK received before the fallback cancels that redundant transmission. A timeout rescue uses a profile-specific fallback radius, while first-contact route-miss recovery remains conservative.

The contributions are:

1) an ACK-aware redundancy controller that separates destination DATA arrival from source-confirmed application delivery;
2) a small state/action/reward formulation that is implementable on MCU-class nodes and changes only a few data-plane parameters;
3) a reproducible seven-configuration comparison covering baselines, ablations, common-random-number scenarios, and stress conditions; and
4) an explicit analysis of the reliability-airtime-delay tradeoff, including the cases where Smart-CALM does not dominate every metric.

II. Problem and Design Goals

A. Reliability Has Two Observable Endpoints

For a unicast flow, the destination can receive DATA even when the source never receives the returning ACK. Reporting only destination arrival would therefore overstate application-level reliability. Let F be the set of unicast flows, D the flows whose DATA reaches the destination, and A the flows whose ACK reaches the source. We report

$$\mathrm{PDR}_{\mathrm{ACK}} = \frac{|A|}{|F|}, \qquad \mathrm{PDR}_{\mathrm{DATA}} = \frac{|D|}{|F|}. \tag{1}$$

The first quantity is the primary application-level metric. The second diagnoses failures in the reverse ACK path and prevents the controller from confusing destination reachability with confirmed delivery.

B. Resource-Constrained Objective

The controller seeks high confirmed delivery while limiting airtime, collisions, route repairs, fallback redundancy, and delay. A useful abstract objective is

$$J = \lambda_d \mathrm{PDR}_{\mathrm{ACK}} - \lambda_a T_{\mathrm{air}} - \lambda_c C - \lambda_r R - \lambda_f F_{\mathrm{fallback}} - \lambda_\ell \bar{L}, \tag{2}$$

where T_{air} is total time on air, C is collision pressure, R is route-repair activity, F_{fallback} is fallback forwarding, and $\bar{L}$ is observed delivery delay. Smart-CALM does not solve this objective globally; it uses a small online approximation to choose among bounded profiles. This keeps the control surface narrow enough for a direct-radio firmware implementation.

III. Smart-CALM Design

A. Forwarding Plane

Smart-CALM uses a four-message data plane: route request (RREQ), route reply (RREP), DATA, and ACK. A source without a valid route floods an RREQ. Candidate paths reaching the destination are retained during a short discovery window; the source then selects the candidate with the highest estimated confidence, breaking ties by hop count. The RREP returns along the reverse path and installs a source-route cache entry.

Subsequent DATA packets carry the complete path. The destination emits an ACK on the reverse path, and the source marks the flow delivered only when that ACK arrives. This is intentionally stricter than counting destination reception. For a path that is weak or stale, Smart-CALM may schedule a second FALLBACK transmission after a delay based on the path airtime. FALLBACK is a bounded managed flood with duplicate suppression. If the source receives the ACK before the delayed fallback begins, the pending fallback is cancelled. This coupling prevents a successful source route from needlessly adding a second flood.

B. Path Confidence

For a received packet, let m be the SNR margin relative to the demodulation threshold and let I_c indicate a collision. The link confidence used during route discovery is

$$c_\ell = \mathrm{clip}\left(\frac{1}{1+\exp(-0.7m)} - 0.18 I_c,\ 0.05,\ 0.98\right). \tag{3}$$

For a candidate path p, the inherited confidence is the minimum hop confidence, with a hop penalty:

$$c(p) = \mathrm{clip}\left(\min_{\ell \in p} c_\ell - 0.025(|p|-2),\ 0.05,\ 0.98\right). \tag{4}$$

Cached routes are aged by a normalized lifetime penalty. The resulting score is used for route selection and for deciding whether a bounded fallback should be scheduled. Confidence is not a guarantee of delivery; it is a local signal that lets redundancy respond to link evidence.

C. Profiles and Timeout Rescue

The controller selects one of three profiles:

- *lean*: 900 s route lifetime, 1-hop timeout rescue, and low path penalties;
- *balanced*: 600 s route lifetime, 2-hop timeout rescue, and the default discovery and aging parameters; and
- *rescue*: 360 s route lifetime, a 2.8 s discovery window, and 3-hop timeout rescue after a low-confidence observation.

The profiles are not separate protocols. They are bounded parameter sets for the same forwarding plane. The route-miss path remains conservative and uses the maximum hop budget unless an explicit experiment option overrides it. The v2 implementation evaluated here applies the selected profile mainly to per-flow route lifetime, confidence handling, and late timeout rescue. This distinction matters because reducing first-contact discovery scope would confound efficiency with reachability.

D. Online Policy

Every 30 s, the controller aggregates a snapshot of recent metrics. The reliability bucket is low, medium, or high based on ACK PDR and route-cache miss ratio. A second bit records whether the collision rate exceeds 0.18, giving six states. The action is one of the three profiles. The window reward is

$$r = 1.6 p_{\mathrm{ACK}} + 0.4 g_{\mathrm{broadcast}} - 0.03\bar{L} - 0.35\rho_{\mathrm{control}} - 0.012\rho_{\mathrm{collision}} - 0.05 N_{\mathrm{repair}} - 0.004 N_{\mathrm{fallback}}. \tag{5}$$

The tabular update is

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)], \tag{6}$$

with $\alpha = 0.45$ and $\gamma = 0.75$ in the default implementation. A small exploration probability is retained, and a weak profile bias favors lean actions when values are tied. The fixed-profile ablation uses the balanced profile without online switching. The no-fallback ablation disables bounded rescue, and the no-confidence ablation selects the shortest route candidate instead of the highest-confidence candidate.

IV. Experimental Method

A. Packet-Level LoRa Model

We use the repository's discrete-event packet-level simulator. Nodes are placed in a 3000 m square. Propagation uses log-distance path loss with log-normal shadowing. The PHY is fixed at SF9, 125 kHz bandwidth, coding rate 4/5, 17 dBm transmit power, and 32-byte payloads. The channel is half-duplex, and overlapping packets fail unless the desired signal exceeds the strongest interferer by a 6 dB capture threshold. Airtime follows the standard LoRa symbol and payload calculation [10]; the radio parameter names and current assumptions are aligned with an SX1262-class device [11].

All protocol runs for a given seed share node placement, traffic schedule, PHY settings, and a static per-link shadowing realization. This common-random- number design reduces variance in protocol differences. The simulator is a research

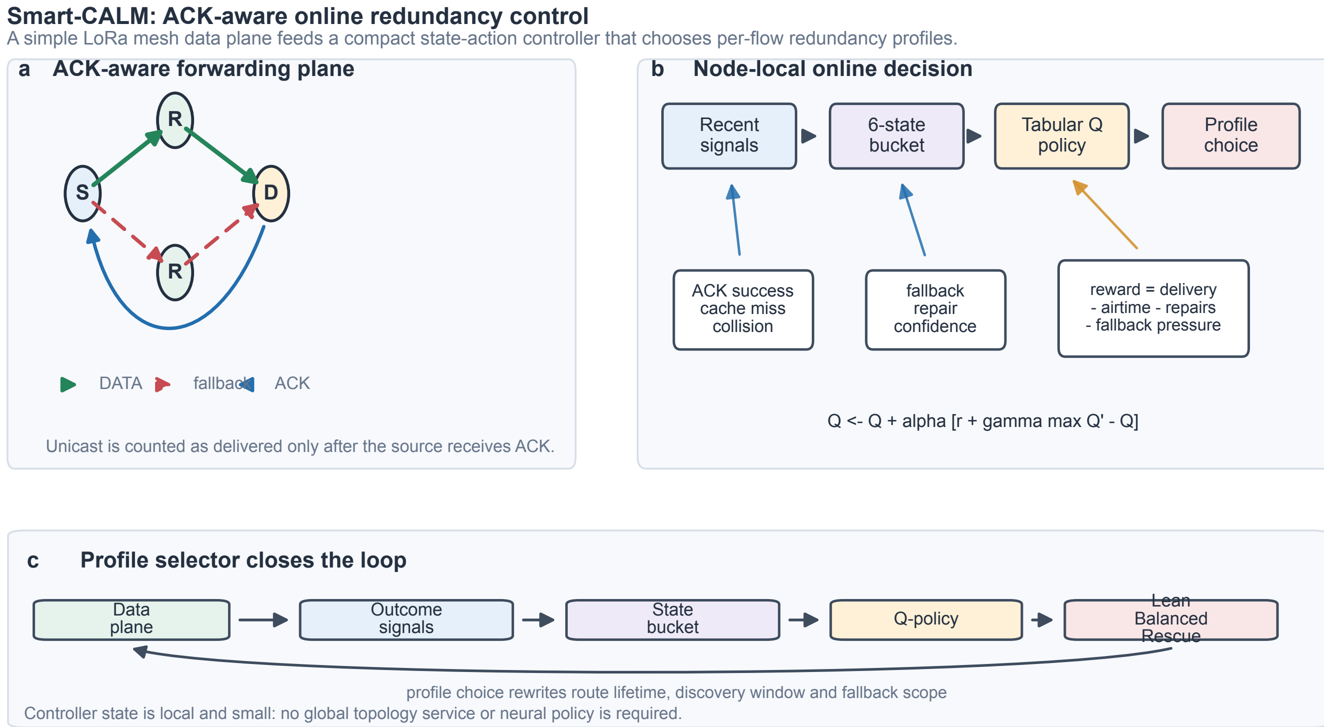

Fig. 1. Smart-CALM architecture. The forwarding plane exposes source-routed DATA, bounded fallback, and reverse-path ACKs. A node-local controller buckets recent outcomes, updates a compact tabular policy, and selects a profile that changes route lifetime, discovery timing, and timeout-rescue scope.

model, not a waveform emulator and not a line-by-line implementation of Meshtastic or MeshCore. The two named baselines are behavior-equivalent models: managed flooding and route discovery plus source-route caching.

B. Comparison Matrix

The matrix contains seven configurations: Meshtastic-like managed flooding, MeshCore-like route caching, nonadaptive CALM, full Smart-CALM, and three Smart-CALM ablations. We use four scenarios, each with 20 seeds:

1) repeated unicast at 6 messages/min;
2) mixed unicast/broadcast traffic at 6 messages/min;
3) mixed traffic with shadowing increased from 4 to 6 dB; and
4) mixed traffic at 10 messages/min.

Each run lasts 1200 s with eight fixed unicast pairs. The resulting raw CSV contains 140 records per scenario (seven configurations times 20 seeds).

C. Metrics and Statistics

For every seed we record the two unicast reliability measures, broadcast coverage, mean and P95 delivery/ACK delay, transmission counts, total airtime, channel busy ratio, energy, packet reception ratio, collision rate, route repairs, fallback forwarding, and policy telemetry. Energy is an accounting model based on the stated transmit and receive currents; it is not a board measurement. For each protocol and metric, we report the mean, sample standard deviation, and a two-sided 95% Student-t interval over the 20 seeds.

TABLE I
DEFAULT SIMULATION CONFIGURATION.

Parameter	Value
Nodes / area	50 / 3000 m $\times$ 3000 m
Duration / seeds	1200 s / 20
Unicast pairs / load	8 / 6 or 10 messages/min
PHY	SF9, BW 125 kHz, CR 4/5
Payload / transmit power	32 bytes / 17 dBm
Path loss / shadowing	2.7 / 4 or 6 dB
Capture threshold	6 dB
Energy model	120 mA Tx, 10.3 mA Rx, 3.3 V

V. Results

A. Mixed-Traffic Main Comparison

Table II shows the main mixed-traffic result. Smart-CALM obtains 0.937 ACK PDR, exceeding managed flooding by 0.085 and CALM by 0.202, while using 29.3% less airtime than managed flooding. Its total energy is close to CALM, but its P95 ACK delay is 14.773 s because the metric includes late rescue and reverse-path confirmation. The result is therefore a reliability gain with a tail-latency cost.

B. Scenario Robustness

Table III compresses the four scenario means for the primary reliability and airtime metrics. Smart-CALM remains

TABLE II
MIXED-TRAFFIC COMPARISON, MEANS OVER 20 SEEDS.

Protocol	ACK PDR	DATA PDR	P95 ACK delay (s)	Airtime (s)	Collision rate	Fallback fwd.	Energy (J)
Meshtastic-like	0.852	0.955	1.119	1182.9	0.526	0.0	2017.2
MeshCore-like	0.389	0.661	0.646	971.5	0.427	0.0	1578.8
CALM	0.735	0.769	2.740	839.6	0.443	50.5	1384.5
Smart-CALM	**0.937**	**0.967**	14.773	**836.6**	0.449	104.2	1388.5
Smart-CALM static	0.941	0.957	15.794	913.6	0.449	357.1	1512.5
Smart-CALM no-fallback	0.768	0.797	2.608	828.6	0.450	64.8	1371.5
Smart-CALM no-confidence	0.937	0.967	14.773	836.6	0.449	104.2	1388.5

above CALM and MeshCore-like on ACK PDR in every scenario. Under stronger shadowing its ACK PDR is 0.928, while managed flooding reaches 0.842. Under the 10 messages/min load, all methods degrade, but Smart-CALM retains 0.871 ACK PDR versus 0.571 for CALM and 0.364 for MeshCore-like. The higher-load result also raises Smart-CALM airtime to 1344.9 s, showing that rescue forwarding spends more channel resources as reliability becomes harder to maintain.

C. Ablation Findings

The ablations help separate the source of the reliability gain. In mixed traffic, disabling fallback reduces ACK PDR from 0.937 to 0.768 while reducing airtime only slightly, which indicates that bounded rescue is doing real recovery work. The static balanced profile reaches a similar PDR but uses 913.6 s of airtime and 357.1 fallback forwards, compared with 836.6 s and 104.2 for the adaptive controller. This supports the claim that profile selection controls redundancy rather than merely enabling a permanently aggressive mode.

The confidence-ranking ablation is currently identical to Smart-CALM in the first three scenarios and differs only slightly at high load. This is not evidence that confidence ranking is useless. It means the sampled topologies rarely contain a strong conflict between path length and confidence. A targeted topology with two competing routes is required before making a stronger ablation claim.

VI. Discussion and Limitations

A. Interpreting the Tradeoff

Smart-CALM's primary advantage is not a universal reduction in every cost. Compared with managed flooding, it cuts mixed-traffic airtime by 346.4 s while raising ACK PDR by 0.085. Compared with CALM, it raises ACK PDR by 0.202 at almost the same airtime, but its P95 ACK delay is more than five times larger. The mechanism spends this delay on delayed fallback and ACK confirmation. Applications with a strict latency deadline may prefer CALM or the no-fallback variant; applications that value confirmed delivery under unstable paths may prefer Smart-CALM.

B. Reproducibility and Scope

The complete matrix is generated by a shell script and stored as raw per-seed CSV plus long-format summary CSV. The simulator uses fixed random seeds and common propagation realizations across protocols. These controls make the comparison auditable, but they do not turn a packet-level model into a physical LoRa testbed. The energy values use 120 mA transmit current, 10.3 mA receive current, and 3.3 V; board startup, sleep current, antenna mismatch, and channel-regulatory duty-cycle effects are outside the model.

The evaluated scale is 50 nodes in a square area. The results do not establish scalability to hundreds of nodes, dense interference environments, or changing traffic beyond the tested load. The baselines are behavior-equivalent models, not claims of compatibility with the Meshtastic or MeshCore wire protocols. Finally, the confidence-ranking ablation needs a deliberately constructed route-conflict scenario, and the firmware prototype still requires board-level radio validation before deployment claims are warranted.

VII. Related Work

LoRa and LoRaWAN surveys describe the sensitivity of long-range low-rate systems to airtime, gateway reachability, and multihop design [1], [2], [12]. Mesh extensions have explored distance-vector routing, clustering, and compact implementation strategies [5], [6], [9]. Smart-Hop and MRT-LoRa address latency and real-time forwarding concerns [7], [8]. Smart-CALM differs in its control boundary: it keeps the forwarding plane explicit and uses ACK-confirmed outcomes to choose a bounded redundancy profile at runtime. Its path confidence is also deliberately local and lightweight, related in spirit to link-quality path metrics such as ETX [13], but with an explicit penalty for route age and a separate ACK endpoint.

VIII. Conclusion

This paper introduced Smart-CALM, an ACK-aware online redundancy controller for airtime-constrained LoRa mesh networks. Across four 20-seed scenarios, the method improved source-confirmed delivery over managed flooding, route-cache forwarding, and nonadaptive CALM while staying below flooding airtime in the main mixed workload. The results also exposed a meaningful cost: longer tail ACK delay and more fallback activity. Treating this as a three-way reliability-airtime-delay tradeoff gives a more defensible contribution than claiming metric dominance. The next evaluation step is a targeted route conflict scenario for the confidence ablation, followed by waveform- or hardware-level validation of the packet-level findings.

TABLE III
PRIMARY METRICS ACROSS ALL SCENARIOS, MEANS OVER 20 SEEDS.

Scenario	Protocol	ACK PDR	DATA PDR	P95 ACK delay (s)	Airtime (s)	Collision rate	Energy (J)
Repeated unicast	Meshtastic-like	0.838	0.949	1.769	1152.0	0.526	1976.6
Repeated unicast	MeshCore-like	0.605	0.812	0.592	357.8	0.388	603.0
Repeated unicast	CALM	0.831	0.856	2.740	256.3	0.345	438.0
Repeated unicast	Smart-CALM	**0.936**	**0.961**	14.876	289.6	0.351	498.5
Mixed traffic	Meshtastic-like	0.852	0.955	1.119	1182.9	0.526	2017.2
Mixed traffic	MeshCore-like	0.389	0.661	0.646	971.5	0.427	1578.8
Mixed traffic	CALM	0.735	0.769	2.740	839.6	0.443	1384.5
Mixed traffic	Smart-CALM	**0.937**	**0.967**	14.773	**836.6**	0.449	1388.5
Shadowing 6 dB	Meshtastic-like	0.842	0.949	1.236	1175.1	0.529	2008.1
Shadowing 6 dB	MeshCore-like	0.387	0.662	0.622	969.6	0.428	1575.7
Shadowing 6 dB	CALM	0.731	0.775	2.740	833.8	0.447	1378.3
Shadowing 6 dB	Smart-CALM	**0.928**	**0.960**	14.673	842.6	0.445	1395.8
Mixed, 10/min	Meshtastic-like	0.714	0.914	2.037	1876.9	0.546	3234.0
Mixed, 10/min	MeshCore-like	0.364	0.530	0.945	1442.7	0.448	2368.5
Mixed, 10/min	CALM	0.571	0.633	2.740	1266.8	0.470	2112.6
Mixed, 10/min	Smart-CALM	**0.871**	**0.920**	18.199	1344.9	0.469	2254.2

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